

Disproof of Conjecture 00000007130: the period of an Ehrhart quasi-polynomial need not be the lcm of the denominators

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Abstract

Conjecture 00000007130 asserts that the period of the Ehrhart quasi-polynomial of a rational polytope is the least common multiple of its denominators. This is false. The triangle $T = \text{conv}\{(0, 0), (1, \frac{1}{2}), (2, 0)\}$ has vertex denominators with lcm 2, yet $\#(tT \cap \mathbb{Z}^2) = \frac{(t+1)(t+2)}{2}$ for every integer $t \geq 0$. So its Ehrhart quasi-polynomial is a polynomial and its period is 1. This is the smallest case of the *period collapse* of McAllister and Woods. If “denominators” is read as the denominators of the facet inequalities instead, the triangle $T_2 = \text{conv}\{(0, 0), (1, 0), (0, \frac{1}{2})\}$ refutes the claim: its facets are $x \geq 0$, $y \geq 0$, $x + 2y \leq 1$ (lcm 1), but its Ehrhart quasi-polynomial has period exactly 2. The second counterexample also defeats the weaker reading “the lcm is a period”. Both results are proved in Lean 4 (core library only), with the polytopes, lattice-point counts, quasi-polynomials and periods defined from scratch.

1 The conjecture and its readings

The statement (English version) reads:

Definition: Integer programming with lattice polytopes. *Conjecture:* The period of the Ehrhart quasi-polynomial is the least common multiple of denominators, and the explicit form is the inverse normalization.

The Chinese version says the same. The conjecture is a conjunction. We refute its first clause:

$$\text{the period of the Ehrhart quasi-polynomial of } P = \text{lcm}(\text{denominators of } P). \quad (1)$$

We do not try to interpret the second clause (“the explicit form is the inverse normalization”). A conjunction with a false conjunct is false.

Lattice versus rational polytopes. For a lattice polytope (integer vertices) the Ehrhart function is a polynomial (Ehrhart’s theorem) and every denominator is 1. Then (1) says “ $1 = 1$ ” and has no content. The words “quasi-polynomial”, “period” and “denominators” only mean something for *rational* polytopes, whose vertices have rational coordinates. So the only reading in which the conjecture says anything is the standard rational setting: $P \subset \mathbb{R}^d$ a rational polytope and $L_P(t) = \#(tP \cap \mathbb{Z}^d)$. We adopt this reading. If the conjecture is meant to speak only about lattice polytopes, it is a (trivially) true statement and there is nothing to refute. We state this limitation openly.

“The period”. A quasi-polynomial has many periods (every multiple of a period is a period). With the definite article, “the period” is the *minimal* period. This is how it is used in the literature, e.g. McAllister–Woods, *The minimum period of the Ehrhart quasi-polynomial of a rational polytope* [3]. If the clause only said “the lcm is a period”, then in the vertex reading it would be Ehrhart’s theorem, which is true (Remark 5). Under the facet reading even this weak form fails (Section 4).

“Denominators”. We treat three readings.

- (V) *Vertex denominators* (standard): the lcm of the denominators of all vertex coordinates, i.e. the least $d \geq 1$ such that dP is a lattice polytope. Refuted by T (Section 3).
- (F) *Facet denominators*: write each facet inequality as $a \cdot x \leq b$ with a primitive integer normal a and $b \in \mathbb{Q}$ in lowest terms, and take the lcm of the denominators of the b ’s. Equivalently, it is the least d such that every facet hyperplane of dP contains a lattice point. Refuted by T_2 (Section 4). This holds for both “the period” and “a period”.
- (C) *Coefficient denominators*: the lcm of the denominators of the coefficients of the quasi-polynomial. Refuted by T (Remark 6).

Any reading that takes the lcm over vertex *and* facet denominators gives 2 for T , and is also refuted by T .

2 Definitions

Definition 1. A function $L : \mathbb{Z}_{\geq 0} \rightarrow \mathbb{Z}$ is a *quasi-polynomial with period* $p \geq 1$ if there are polynomials $c_0, \dots, c_{p-1} \in \mathbb{Q}[t]$ with $L(t) = c_{t \bmod p}(t)$ for all $t \geq 0$. The (*minimal*) *period* of L is the least such p . It is unique, and every multiple of a period is a period.

In Lean the rational polynomials are written over a common denominator: $D \cdot L(t) = P_{t \bmod p}(t)$ with $D \in \mathbb{Z}_{>0}$ and $P_r \in \mathbb{Z}[t]$. This is equivalent.

For a rational polytope P , Ehrhart’s theorem [1] says that $L_P(t) = \#(tP \cap \mathbb{Z}^d)$ is a quasi-polynomial and that the vertex-denominator lcm is a period. McMullen [2] refined this coefficient by coefficient. McAllister and Woods [3] showed that the minimal period can be strictly smaller (“period collapse”). Their family $\text{conv}\{(0,0), (1, \frac{D-1}{D}), (D,0)\}$ has denominator D and period 1, and our T is the case $D = 2$. We give a complete self-contained proof for T below.

3 The counterexample $T = \text{conv}\{(0,0), (1, \frac{1}{2}), (2,0)\}$

The vertex coordinates $0, 0, 1, \frac{1}{2}, 2, 0$ are in lowest terms with denominators $1, 1, 1, 2, 1, 1$, so

$$\text{lcm}(\text{vertex denominators of } T) = 2.$$

Lemma 2 (lattice points of tT). *Let $t \geq 0$ and $(x, y) \in \mathbb{Z}^2$. Then $(x, y) \in tT$ if and only if*

$$y \geq 0, \quad 2y \leq x, \quad x + 2y \leq 2t.$$

All such points lie in the box $0 \leq x \leq 2t, 0 \leq y \leq t$.

Proof. $(x, y) \in tT$ means $(x, y) = \nu_0(0, 0) + \nu_1(1, \frac{1}{2}) + \nu_2(2, 0)$ with $\nu_i \geq 0$ rational and $\nu_0 + \nu_1 + \nu_2 = t$. (For $t = 0$ this gives $\{(0, 0)\} = 0T$.) The vertices are affinely independent, so the weights are unique: $y = \nu_1/2$ and $x = \nu_1 + 2\nu_2$ give

$$\nu_1 = 2y, \quad \nu_2 = \frac{x-2y}{2}, \quad \nu_0 = t - \nu_1 - \nu_2 = \frac{2t-x-2y}{2}.$$

Nonnegativity of these is exactly the three inequalities. Conversely, if they hold, these ν_i work. For the box: $0 \leq 2y \leq x \leq 2t - 2y \leq 2t$ and $4y \leq x + 2y \leq 2t$. $\square$

The three inequalities are the facets $-y \leq 0$, $-x + 2y \leq 0$, $x + 2y \leq 2$ of T , scaled by t . Their normals are primitive and their right-hand sides are integers, so the facet lcm (F) of T is 1.

Theorem 3 (period collapse). *For every integer $t \geq 0$,*

$$L_T(t) = \#(tT \cap \mathbb{Z}^2) = \frac{(t+1)(t+2)}{2}.$$

Proof. By Lemma 2, row y ($0 \leq y \leq t$) contains the integers x with $2y \leq x \leq 2t - 2y$. That is $\max(0, 2t + 1 - 4y)$ points. If $4y \leq 2t$ this is $2t + 1 - 4y$. If $4y > 2t$ then $4y \geq 2t + 2$, so the row is empty. Hence $L_T(t) = \sum_{0 \leq y \leq t/2} (2t + 1 - 4y)$.

- $t = 2m$: $\sum_{y=0}^m (4m + 1 - 4y) = (m+1)(4m+1) - 2m(m+1) = (m+1)(2m+1) = \frac{(t+1)(t+2)}{2}$.
- $t = 2m+1$: $\sum_{y=0}^m (4m+3-4y) = (m+1)(4m+3) - 2m(m+1) = (m+1)(2m+3) = \frac{(t+1)(t+2)}{2}$.

(The Lean proof uses the equivalent recurrence $L_T(t+2) = L_T(t) + 2t + 5$. Row $y+1$ of $(t+2)T$ has as many points as row y of tT , and row 0 of $(t+2)T$ adds $2t+5$. Together with $L_T(0) = 1$ and $L_T(1) = 3$ this gives the formula by induction.) $\square$

Corollary 4. *The Ehrhart quasi-polynomial of T has minimal period 1, while the lcm of the vertex denominators of T is 2. Hence clause (1) fails for T in reading (V).*

Proof. Theorem 3 gives period 1 with $c_0(t) = \frac{1}{2}t^2 + \frac{3}{2}t + 1$, and no period is smaller than 1. $\square$

Remark 5 (“a period”). The number 2 *is* a period of L_T (take $c_0 = c_1$), as Ehrhart’s theorem guarantees. So the counterexample needs “the period” to mean the minimal period. That is the meaning of the definite article and the meaning in [3]. In Lean this is `LT.period_denLcm`.

Remark 6 (coefficient denominators). The Ehrhart polynomial $\frac{1}{2}t^2 + \frac{3}{2}t + 1$ has coefficient denominators 2, 2, 1, with lcm 2 $\neq$ 1. So reading (C) is refuted by T too.

Remark 7. The same happens for every member of the McAllister–Woods family. `verify.py` checks $D = 2, \dots, 6$: denominator D , minimal period 1.

4 The facet reading: $T_2 = \text{conv}\{(0, 0), (1, 0), (0, \frac{1}{2})\}$

T_2 has facets $-x \leq 0$, $-y \leq 0$, $x + 2y \leq 1$. Each has a primitive integer normal and an integer right-hand side, so the lcm in reading (F) is 1. Each facet line passes through two vertices. Its vertex denominators have lcm 2.

Lemma 8. *For $t \geq 0$ and $(x, y) \in \mathbb{Z}^2$: $(x, y) \in tT_2$ iff $x \geq 0$, $y \geq 0$, $x + 2y \leq t$. Such points lie in $[0, t]^2$.*

Proof. The unique weights are $\nu_1 = x$ (for $(1, 0)$), $\nu_2 = 2y$ (for $(0, \frac{1}{2})$) and $\nu_0 = t - x - 2y$. $\square$

Theorem 9. $4L_{T_2}(t) + (t \bmod 2) = (t + 2)^2$ for all $t \geq 0$. So $L_{T_2}(t) = \frac{(t+2)^2}{4}$ for even t and $\frac{(t+1)(t+3)}{4}$ for odd t , and the minimal period of L_{T_2} is exactly 2.

Proof. Row y has $\max(0, t + 1 - 2y)$ points, so $L_{T_2}(t) = \sum_{0 \leq y \leq t/2} (t + 1 - 2y)$. For $t = 2m$ this is $(m + 1)^2$, and for $t = 2m + 1$ it is $(m + 1)(m + 2)$. (Lean: $L(t + 2) = L(t) + t + 3$.) So 2 is a period.

Suppose L_{T_2} had period 1, i.e. $L_{T_2}(t) = q(t)$ for all $t \geq 0$ with $q \in \mathbb{Q}[t]$. Then $q(t) = (t + 2)^2/4$ for infinitely many (even) t , so $q = (t + 2)^2/4$. But $q(1) = 9/4 \neq 2 = L_{T_2}(1)$. The Lean proof uses an integral version of this argument. Write $D \cdot L_{T_2} = P$ with $D \geq 1$ and $P \in \mathbb{Z}[t]$, and put $a = 2D + 1$. Then $a \mid P(a) - P(0) = D(L_{T_2}(a) - 1)$. Now $4D(L_{T_2}(a) - 1) = D(a^2 + 4a - 1) = Da(a + 4) - D$, so $a \mid D$. This contradicts $0 < D < a$. $\square$

Corollary 10. In reading (F), clause (1) fails for T_2 : the facet lcm is 1 and the minimal period is 2. Since 1 is not even a period, the weak form fails too.

Remark 11. Each triangle satisfies the clause under the *other* reading: T has facet lcm 1 = period, and T_2 has vertex lcm 2 = period. Lean proves both (`sanity_T_facet`, `sanity_T2_vertex`). This shows the formal claims are not vacuous: the predicates can hold. Taken together, the two triangles refute every reading (V), (F), (C), and the combined vertex-and-facet lcm.

5 Formalization in Lean 4

The project `lean4/` (Lean 4.19.0, core library only) defines, from scratch:

- `RQ`, `RPt`, `Tri`: rational numbers `num/den`, rational points, triangles by vertices. `Tri.Reduced` means the coordinates are in lowest terms. `Tri.denLcm` is the lcm of the six vertex denominators.
- `Tri.MemScaled P t x y`: $(x, y) \in tP$. There are $M > 0$ and $\mu_i \geq 0$ with $\sum \mu_i = Mt$ and $(x, y) = \sum (\mu_i/M)v_i$, with fractions cleared by M and by the product of the denominators.
- `Facet`: $a_1x + a_2y \leq b$ with $b \in \mathbb{Q}$. `facetDenLcm`, `Facet.Primitive` and `FacetsOf P fs` (each facet is primitive and its line passes through two vertices).
- `hcount fs W H t`: the number of $(x, y) \in \mathbb{Z}^2$ with $0 \leq x \leq Wt$, $0 \leq y \leq Ht$ that satisfy every facet inequality scaled by t , computed by explicit finite counting. `Describes P fs W H` states that for all t, x, y , `MemScaled P t x y` holds iff (x, y) is in the box and satisfies the facets. Then `hcount fs W H` is exactly L_P .
- `peval` (integer polynomials as coefficient lists), `HasPeriod L p` and `IsMinPeriod L p`, with `isMinPeriod_unique`.

Lean theorem	content
<code>describes_T, describes_T2</code>	Lemmas 2, 8 (V- and H-description agree)
<code>T_denLcm, Tfacets_denLcm</code>	vertex lcm 2, facet lcm 1 for T
<code>LT_formula</code>	$2L_T(t) = (t+1)(t+2)$
<code>LT_minPeriod, LT_not_minPeriod_two</code>	minimal period of L_T is 1, not 2
<code>LT_period_denLcm</code>	2 is a period of L_T
<code>L2_formula, L2_minPeriod</code>	$4L_{T_2}(t) + (t \bmod 2) = (t+2)^2$; minimal period 2
<code>conjecture_00000007130_false</code>	$\neg \text{VertexClaim}$
<code>conjecture_00000007130_false_facet</code>	$\neg \text{FacetClaim}$
<code>conjecture_00000007130_false_facet_weak</code>	$\neg \text{FacetClaimWeak}$ (“a period”)

VertexClaim says: for every rational triangle P in lowest terms, and every facet list and box that describe it, `IsMinPeriod (hcount fs W H) P.denLcm`. **FacetClaim** is the same with `facetDenLcm fs` and the extra hypothesis `FacetsOf P fs`. These are claims over all rational triangles, and triangles are rational polytopes, so their negations refute the clause for rational polytopes. There is no `sorry`, no `native.decide` and no added axiom. `#print axioms` reports only `propext`, `Classical.choice` and `Quot.sound`.

Limitations. Reading (C) is treated only in this report and in `verify.py`. The Lean formalization works with triangles and with an explicit facet list plus box. Their agreement with the vertex description is proved, not assumed (`describes_T, describes_T2`). The literal “lattice polytope” reading is not refuted, because the clause is trivially true there.

6 Independent check

`verify.py` uses only the Python standard library. It counts lattice points of tP directly from the *vertices*, using exact barycentric coordinates (Cramer’s rule with fractions) on a generous box. It derives the primitive facet inequalities from the vertices. It determines the minimal period by exact interpolation of each residue class with polynomials of degree ≤ 2 , for all $t \leq 60$. Results: $L_T(t) = (t+1)(t+2)/2$, with period 1, vertex lcm 2, facet lcm 1, coefficient lcm 2. L_{T_2} has period 2, vertex lcm 2, facet lcm 1. The McAllister–Woods triangles with $D = 2..6$ all have period 1.

References

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